

# Demand Analysis of DPV Participating in Power Balance with Consideration of Prediction Confidence Intervals

Ruoying Yu  
Renewable energy Research Center,  
China-Electric Power Research  
Institute  
Nanjing, China  
yuruoying@epri.sgcc.com.cn

Jiaqi Liao  
Renewable energy Research Center,  
China-Electric Power Research  
Institute  
Nanjing, China  
liaojujiaqi@epri.sgcc.com.cn

Rongbo Ye  
Renewable energy Research Center,  
China-Electric Power Research  
Institute  
Nanjing, China  
yerongbo@epri.sgcc.com.cn

Baoju Li  
State Grid, Jilin Electric Power Co., Ltd,  
Changchun, china  
libj02@jl.sgcc.com.cn

Guanqun Zhuang  
State Grid, Jilin Electric Power Co., Ltd,  
Changchun, china  
zhuanggq@jl.sgcc.com.cn

**Abstract**—The explosive development of distributed photovoltaics brings risks to the balance of the power grid. In this paper, a quantitative method of DPV participation in power balance is proposed. Based on the prediction confidence analysis, a DPV participation balance analysis method is constructed, which fully considers the photovoltaic power fluctuation, prediction bias and system operating boundary. The result is a set of time series variables affected by the system operating state and photovoltaic prediction and fluctuation, which can provide theoretical support for the power system scheduling plan preparation and operation. A numerical example shows the effectiveness of the proposed method.

**Keywords**—Demand Analysis; Prediction Confidence; Power balance; Peak regulation; Distributed photovoltaic

## I. INTRODUCTION

With the integration scale of DPV (distributed photovoltaic) getting larger, there is a trend of "distribution network extending to the main network"<sup>[1]</sup>, which has exceeded the scope of local balance. It is hard for some provinces to meet the system's peaking demand, even after abandoning all the centralized renewable energy generated during the midday photovoltaic peak period. DPV faces being forced to abandon power. It is urgently needed for DPV to shoulder the obligation of balance and regulation. In recent years, a lot of research on DPV has mainly focused on its operation control, inverter control and power quality improvement in local distribution networks and micro-grids<sup>[2]-[5]</sup>. However, there is no mature method for quantitative calculation of DPV participation in grid power and energy balance regulation. It is difficult to quickly and accurately evaluate the future power and energy balance regulation of DPV. Therefore, it is urgently needed to develop a method for analyzing the demand for DPV to participate in grid power and energy balance regulation, which can provide a critical foundation and principle for the subsequent power system dispatching, as well as for the DPV participation in the power market.

Under the above background, this paper provides an analysis method of active power demand for DPV participation in power grid balance, which is a set of timing variable that varies with the operating state of the system and distributed photovoltaic power generation. The method is divided into four parts, including analysis of prediction confidence, calculate the proportion of DPV participating in power balance, optimize the startup mode of the generator,

calculate the demand of DPV participating in the electric power balance. The overall description is as follows.

## II. DEMAND ANALYSIS METHOD OF DPV PARTICIPATING IN POWER GRID BALANCE REGULATION

### A. Analysis of Prediction Confidence

The prediction error<sup>[6][7]</sup> analysis of DPV is the basis of this method. It is necessary to analyze the confidence interval of prediction accuracy in different output simultaneity rate intervals in historical power generation data (including measured power and predicted power data). According to the size of the output simultaneity of DPV, the simultaneity interval is divided according to a certain step size as  $[r_1, r_2], [r_2, r_3], \dots, [r_{n-1}, r_n], [r_n, r_{n+1}]$ . The power prediction accuracy of DPV at each time in different output simultaneity intervals is calculated. Specifically, there are four steps

- *Step 1:* Calculate the simultaneity of DPV output at each moment  $r(t)$ ,

$$r(t) = \frac{P_r(t)}{S} \times 100\% \quad (1)$$

$P_r(t)$ —DPV power generation at  $t$ ;

$S$ —DPV installed capacity.

- *Step 2:* According to the output simultaneity distribution interval of DPV, the step size of the simultaneous rate interval is selected as  $\Delta t$ . Thus The DPV simultaneity interval is divided into simultaneity subintervals as  $[r_1, r_2], [r_2, r_3], \dots, [r_{n-1}, r_n], [r_n, r_{n+1}]$ ,

$$\begin{cases} r_1 = \min\{r(t)\} \\ r_2 = r_1 + \Delta r \\ \vdots \\ r_n = r_{n-1} + \Delta r \end{cases} \quad (2)$$

- *Step 3:* Calculate the prediction accuracy of DPV power at each moment, the calculation formula is defined as:

$$\eta(t) = 1 - \frac{|P_{pre}(t) - P_r(t)|}{P_{pre}(t)} \times 100\% \quad (3)$$

$P_{pre}(t)$ ——Predicted value of DPV power generation at  $t$ ;

$\eta(t)$  —— Prediction accuracy of DPV power generation at  $t$ ;

$P_r(t)$ ——Actual generating of DPV power generation at  $t$ ;

- *Step 4:* For each simultaneity interval, the confidence interval of DPV prediction accuracy meeting the requirement of set ratio confidence should be calculated. Take the setting ratio of 90% and the rate interval  $[r_n, r_{n+1}]$  as an example to illustrate the specific calculation process: Select the output prediction accuracy corresponding to the simultaneous rate falling in the  $[r_n, r_{n+1}]$  interval, forming the set  $\{\eta_1^n, \eta_2^n, \dots, \eta_k^n\}$ . They are sorted in order from largest to smallest to form a sequence  $\{\eta_1^n, \eta_2^n, \dots, \eta_k^n\}$ . When the confidence requirement is set to 90%, the prediction accuracy confidence interval is  $\{\eta_1^n, \eta_2^n, \dots, \eta_m^n\}$ ,  $m = [90\% * k]$  (function  $y=f(x)=[x]$  is an integral function, representing the largest integer not exceeding  $x$ ).

According to the above method, the confidence interval of DPV prediction accuracy under the 90% confidence requirement corresponding to each simultaneity interval is calculated, and the calculation results under a total of  $n$  simultaneity rate intervals are obtained as (4):

$$[\eta_1^1, \eta_{m1}^1], [\eta_1^2, \eta_{m2}^2], \dots, [\eta_1^n, \eta_{mn}^n] \quad (4)$$

#### B. Calculate the proportion of DPV participating in power balance

According to the DPV power prediction confidence interval obtained in A, it is able to determine the proportion of distributed PV classification included in the power balance calculation.

Using the predicted value of DPV power, the simultaneity of DPV output is calculated. Then the interval of its simultaneity is determined according to A. The confidence interval of DPV prediction accuracy that meets certain confidence requirements (such as 90%) is determined according to each simultaneity interval obtained, and the confidence interval of DPV power prediction value is determined. At this time, the proportion of DPV included in the power supply balance is determined as the lower limit of the confidence interval, and is denoted as  $\eta_{dpv}$ .

#### C. Optimize the startup mode of the generator

According to the proportion of DPV included in the power balance obtained in B, the startup mode of conventional power units considering DPV could be calculated. Based on the prediction results of DPV power, combined with the pre-built power startup model, the optimization algorithm is used to calculate the power startup mode of the unit. The optimization model of power unit startup mode is constructed as follows:

*Objective function:*

$$F = \min \sum_{i=1}^{N_g} S_i^g \cdot z_i^g \quad (5)$$

$S_i^g$  ——Capacity of conventional power unit  $i$ ;

$i$  ——Serial number of conventional power unit;

$z_i^g$  ——The status of conventional power unit  $i$ , boolean variable, 1 is on, 0 is off.

*Constraint condition*

(1) Positive reserve constraint

$$\sum_{i=1}^{N_g} S_i^g \cdot z_i^g \geq \max \{P_{load} + Re_p - \eta_{dpv}\} \quad (6)$$

$P_{load}$  ——daily load forecast curve;

$Re_p$  —— system's positive reserve request;

$\eta_{dpv}$  ——the proportion of DPV generation at the same time as the load is included in the balance.

(2) Negative reserve constraint

$$\sum_{i=1}^{N_g} S_i^g \cdot z_i^g \cdot \lambda_i^g \leq \min \{P_{load} - Re_n - \eta_{dpv}\} \quad (6)$$

$\lambda_i^g$  —— Minimum technical output of conventional unit  $i$ ;

$Re_n$  —— negative reserve request of system;

(3) Start-stop constraint

$$\begin{cases} (T_{i,on}^{t-1} - UT_i)(z_i^{t-1} - z_i^t) \geq 0 \\ (T_{i,off}^{t-1} - DT_i)(z_i^t - z_i^{t-1}) \geq 0 \end{cases} \quad (7)$$

$T_{i,on}^{t-1}$  ——Unit  $i$  Continuous start-up time;

$T_{i,off}^{t-1}$  ——Unit  $i$  Continuous shutdown time;

$UT_i$  ——The minimum available time of unit  $i$ ;

$DT_i$  ——The minimum off-time of unit  $i$ ;

$z_i^t$  ——State of unit  $i$ , 1 indicates on, 0 indicates power off.

#### D. Calculate the demand of DPV participating in the electric power balance

Based on the annual production simulation sequential operation data, including minimum technical output of conventional power supply, incoming (fed out) of liaison line, power load, output of centralized renewable energy and DPV of provincial power system obtained in C, the demand for DPV participating in the electric power balance could be calculated as following:

Power balance calculation is carried out with the production simulation time series operation data within the annual cycle. The formula (8) is built to be used to determine whether DPV need to participate in electric power balance. If the provincial power system at  $t$  satisfies formula (8), then it means that DPV can be fully absorbed at  $t$ , and the provincial

power system still has absorption space, which could be calculated with formula (9). Else, it means that DPV cannot be fully absorbed at  $t$ , thus the provincial power grid needs DPV to participate in electric power balance. The peak regulation demand of DPV could be obtained with formula (10). The diagram of DPV participating in power balance is shown in the figure1:

$$P_{\text{load}}(t) \geq P_u(t) + P_i(t) + P_{\text{dpv}}(t) + (1 - k_{\text{cpv}})P_{\text{cpv}}(t) \quad (8)$$

$$P_{\text{consum}}(t) = P_{\text{load}}(t) - P_u(t) - P_i(t) - P_{\text{dpv}}(t) - (1 - k_{\text{cpv}})P_{\text{cpv}}(t) \quad (9)$$

$$P_{\text{peak}}(t) = P_u(t) + P_i(t) + P_{\text{dpv}}(t) + (1 - k_{\text{cpv}})P_{\text{cpv}}(t) - P_{\text{load}}(t) \quad (10)$$

$P_{\text{consum}}(t)$ ——DPV consuming space at  $t$ ;

$P_{\text{load}}(t)$ ——Load of Provincial power system at  $t$ ;

$P_u(t)$ ——Minimum technical output of thermal at  $t$ ;

$P_i(t)$ ——Incoming power of liaison line at  $t$ ;

$P_{\text{dpv}}(t)$ ——Output of DPV at  $t$ ;

$P_{\text{cpv}}(t)$ ——Output of centralized renewable energy at  $t$ ;

$K_{\text{cpv}}$ ——maximum abandonment ratio of centralized renewable energy;

$P_{\text{peak}}(t)$ ——The peak regulation demand of DPV at  $t$ .

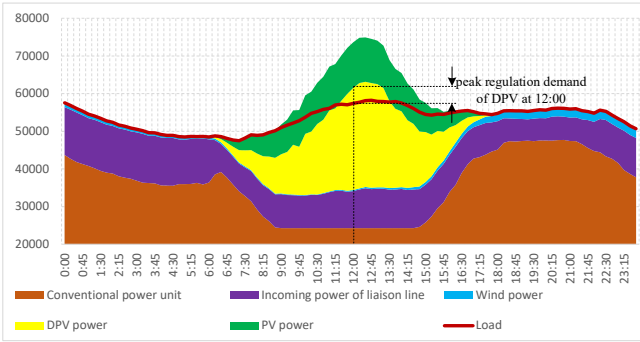


Figure.1 Diagram of DPV participating in power balance

The method above could be applied to analyze the annual operation of provincial power system, calculating the demand for DPV participating in peak regulation with renewable energy and load fluctuations. Results provide a foundation for annual operation mode of the provincial power system with large-scale DPV integration.

### III. CASE STUDY

#### A. Basic situation

A provincial power grid is selected as a simulation test case to illustrate the method proposed in this paper. It is predicted that in the maximum load of 2025 would reach 102070.29 MW, which may happen at 11:00 in August. There are 2217.8 MW of minimum technical output of conventional power units, 27889 MW of maximum output of centralized renewable energy and 31203.58 MW of maximum output of DPV installed capacity in the province, whose DPV penetration rate reaches 30.57%. And the maximum power of the income liaison line will be 28070 MW. The maximum abandonment ratio of centralized renewable energy is set as 1.

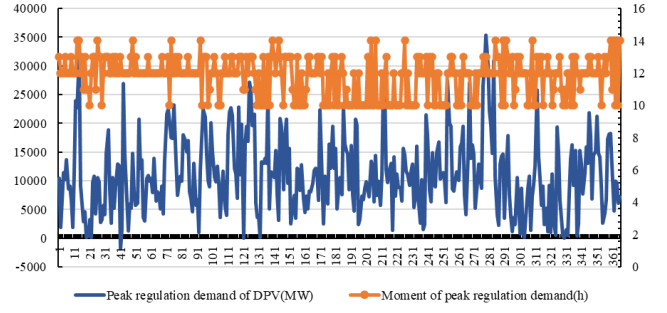


Figure.2 Analysis results of DPV participation in peak shaving demand in a certain province throughout the year

It can be seen that the demand for distributed photovoltaic to participate in peak regulation has occurred in this province, and the peak regulation demand pressure period is mainly concentrated in autumn and winter, while the moment of peak regulation demand is mainly at 12:00, and 10:00 and 13:00 in some seasons.

#### B. The proportion of DPV participating in power balance calculation

This article selects three typical days in winter, spring, and summer to analyze the demand for distributed photovoltaic participation in peak shaving.

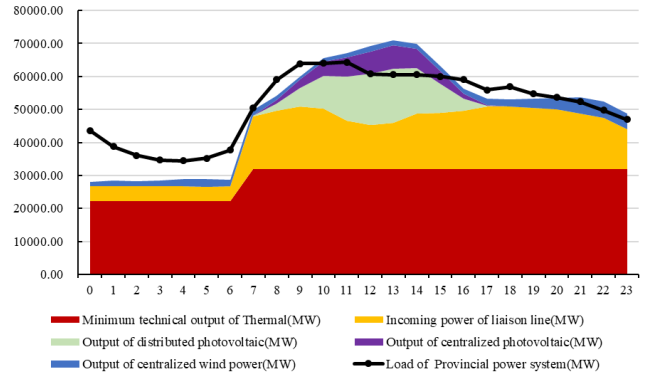


Figure.3 Schematic diagram of provincial power balance on typical winter days

Analyzing the simulated data of the entire day on February 10, 2025, it can be seen from Figure 3 that the output of distributed photovoltaics reaches its maximum value from 11:00 to 14:00, while the load value at 12:00 and 13:00 is already lower than the cumulative curve of the minimum technical output of thermal, the exchange power of interconnection lines, and the output of distributed photovoltaics. In addition, thermal power units need to provide heating, so the minimum technical output has been improved. Moreover, both centralized photovoltaics and centralized wind power are in a state of complete power abandonment. The calculated  $P_{\text{peak}}$  is respectively 44.01 MW and 1788.71 MW, also the maximum value in 2025. Therefore, it can be calculated that the demand of distributed photovoltaics participating in peak shaving in the region in 2025 is 1788.71 MW.

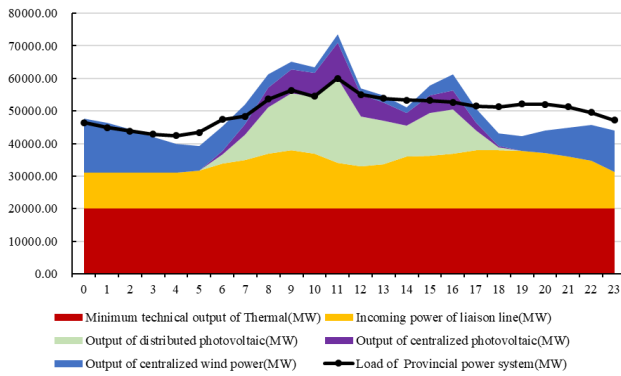


Figure.4 Schematic diagram of provincial power balance on typical spring days

On April 30, 2025, the distributed photovoltaic output reached the maximum value at 11:00, and the cumulative curve of the electricity load was similar to the minimum technical output of conventional power supply, the exchange power of the contact line and the output of distributed photovoltaics.

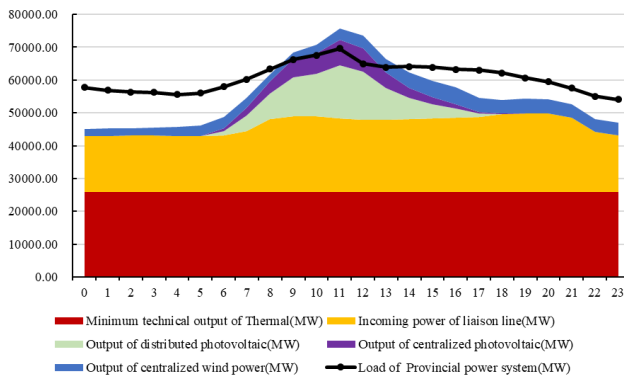


Figure.5 Schematic diagram of provincial power balance on typical autumn days

On October 30, 2025, although the minimum technical output of conventional power supply is higher and the exchange power of the contact line is larger, the distributed photovoltaic output is small due to weather conditions, so the load curve still has a certain amount of room for absorption, which is 190.97MW.

#### IV. CONCLUSION

This paper presents a method for analyzing the demand of DPV participating in power. In the process of integrating DPV into the power balance, the method fully considers the influence of power fluctuation, prediction bias caused by DPV

and system operating boundaries. Confidence interval analysis and optimization modeling method are used. The results calculated is a timing variable that changes in real time with the operation of the system, distributed PV power generation and forecast conditions.

The numerical results show that the method proposed in this paper can accurately quantify the influence of distributed PV on the peak regulation of the power grid and the degree of its participation, effectively solve the pressure brought by large-scale grid-connected distributed PV on the peak regulation of the power grid, and provide a scientific basis for the scheduling operation and mode planning of the power system with a high proportion of new energy access, and improve the level of operation safety.

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